

Cookie Cats Gate Placement

A Product Experimentation Case Study

Product Analytics Case Study • github.com/SambhavyaNayak/cookie-cats-ab-testing-analysis

1. Context

Cookie Cats is a mobile puzzle game where players hit a 'gate' at certain levels, forcing them to wait or pay to continue. The game's first gate sat at level 30. The product team wanted to test whether moving it to level 40 — giving players more uninterrupted play before the first ask — would improve retention. This case study evaluates that experiment for 90,189 players randomly split into two groups (Gate 30 vs. Gate 40) and makes a ship / no-ship recommendation.

2. Hypothesis & Experiment Design

Null hypothesis (H0): Gate placement (30 vs. 40) has no effect on 7-day retention.

Alternative (H1): Gate placement has a measurable effect on 7-day retention.

7-day retention was chosen as the primary metric over 1-day retention because it better reflects sustained engagement rather than a first-session bounce, and it is more predictive of long-term monetization potential.

A statistical power analysis confirmed 88.4% power to detect the observed effect size — meaning the experiment was sufficiently sized to trust a negative result, not just a positive one.

3. Guardrail Check: Was the Test Even Valid?

Before trusting any result, the group split was checked for a Sample Ratio Mismatch (SRM) — a common but often-skipped step that catches broken randomization or logging bugs. The split held at roughly 44,700 (Gate 30) vs. 45,489 (Gate 40) users, consistent with a healthy 50/50 assignment. Without this check, a team risks shipping a decision based on a corrupted experiment.



Fig 1. Executive summary of experiment scale and topline retention deltas.

4. Results: Gate 30 Wins Overall

Gate 30 outperformed Gate 40 on both retention windows. The gap widens over time — a 0.59 point lift at 1 day grows to a 0.82 point lift at 7 days, suggesting the effect compounds rather than fades.

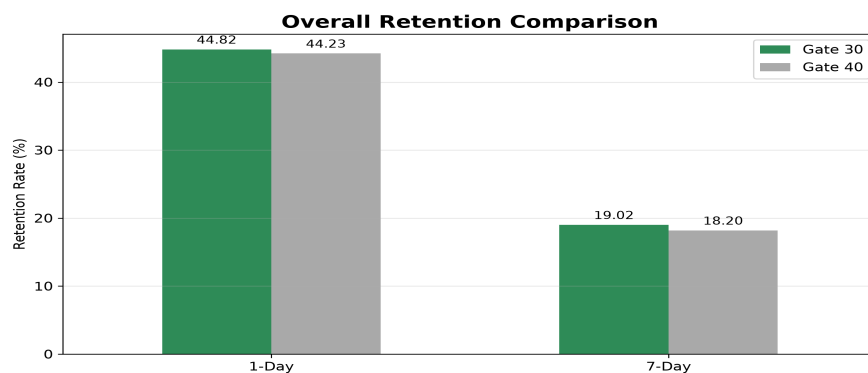


Fig 2. 1-day and 7-day retention, Gate 30 vs. Gate 40.

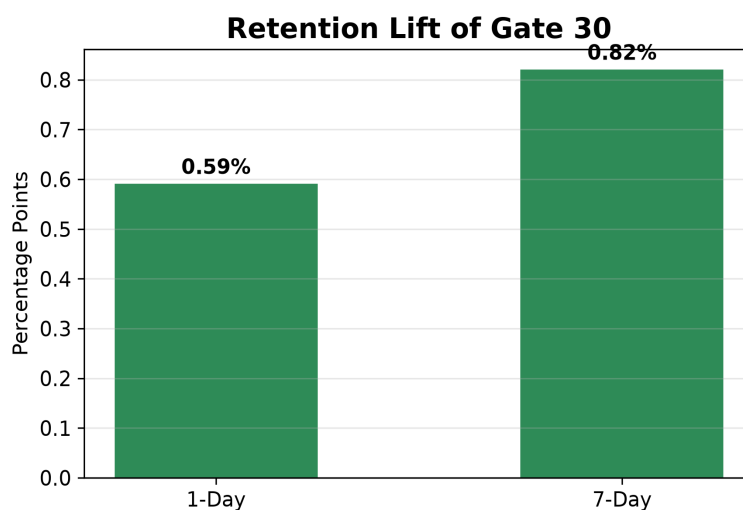


Fig 3. Retention lift of keeping the gate at level 30.

5. The Real Insight: It Depends Who You Ask

Averages hide the story that matters for a product decision. Players were split into Low, Medium, and High engagement segments (roughly equal thirds, ~28-32K players each) based on total game rounds played. The segment breakdown shows the retention drop under Gate 40 is **not evenly distributed**:

- **Low-engagement players:** essentially unaffected, and marginally higher retention under Gate 40 (1.64% vs. 1.54% at day 7).
- **Medium-engagement players:** retention drops from 8.68% to 7.97% at day 7 under Gate 40.
- **High-engagement players:** the largest drop of any segment — 47.66% down to 46.38% at day 7, a 1.28 point loss.

High-engagement players are typically a game's most valuable users — the ones most likely to convert to paying customers over time. Moving the gate later hurts retention most for exactly this group, which a top-line average completely masks.

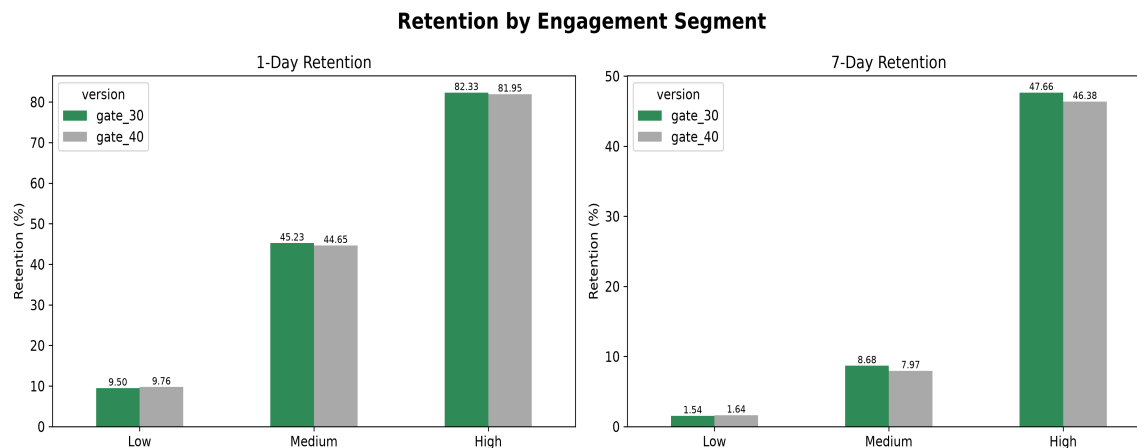


Fig 4. Retention by engagement segment, 1-day and 7-day.

6. Recommendation

Do not move the gate to level 40. Keep it at level 30.

The overall retention lift is small in absolute terms, but it is directionally consistent, grows over the 7-day window, and is concentrated among the game's highest-value players — the segment a product team can least afford to lose. Shipping Gate 40 would trade a marginal early-funnel experience improvement for a measurable retention cost among the most engaged users.

7. Follow-Up Experiment

Rather than treating this as a closed question, the more interesting product move is to test a **soft gate** at level 40 — skippable via a rewarded ad or in-game currency — instead of a hard wall. This could capture the original intent (longer uninterrupted early play) without penalizing high-engagement players who are most sensitive to friction. If shipped, I would monitor 7-day retention by segment, ad-view rate, and in-app purchase conversion as the primary success and guardrail metrics.

Appendix: Player engagement distribution (log scale), used to construct the Low / Medium / High segments above.

